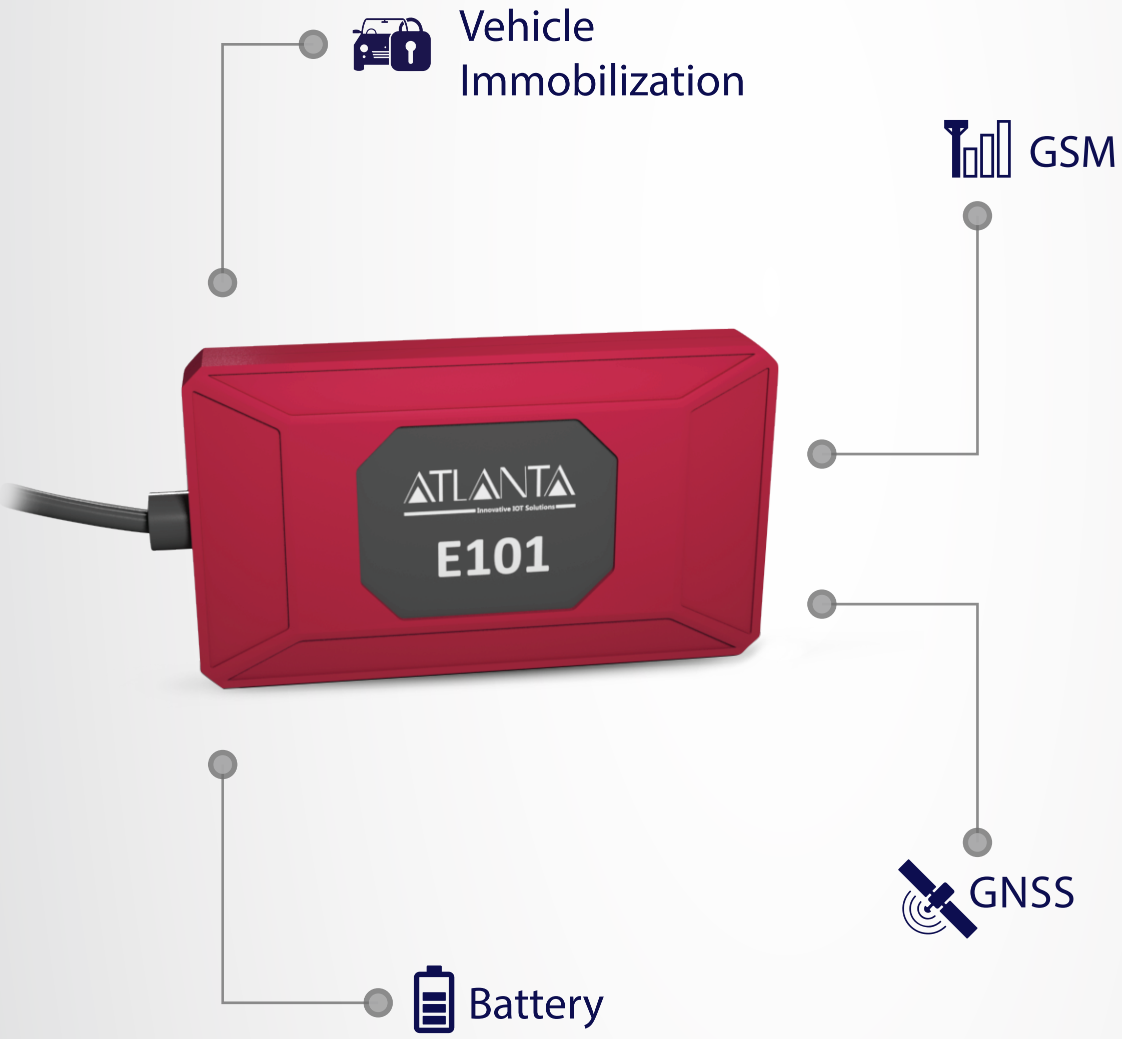
VEHICLE TRACKER

ATLANTA
Innovative IOT Solutions

# E-101

# PROTOCOL

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| SL# | Version | Prepared by | Date |
|---|---|---|---|
| 1 | VTS_GPRS_E101_V1.0 | Vijith V.Nair | 13/03/2021 |

# Protocol

0×20START CHARATERSIGNATUREIMEI,GPS DATA($GPRMC STRINGS) (80 CHAR)$LOC,LOCATION NAME,#I1I2I3I4I5I6I7I8I9I10I11I12I13I14,RESERVED, RESERVED, RESERVED,ODO READING(7Sg),On Board Temp., Internal battery voltage(3Sg),GSM signal strength(%2),MCC(3 digit),MNC(3 digit),LAC(6 char),CellID(5Digits)SIGNATURE END CHARACTER Checksum

## Example :-

- Live data

  0×200×01ATL356895037533745,$GPRMC,111719.000,A,2838.0045,N,07713.3707,E,0.00,,120810,,,A*75$,#01100111001010,6.5,0,0,12345.67,24.4,4.2,21,MCC,MNC,LAC,CellIDATL0×020×7A

- Memory data

  0×200×03ATL356895037533745,$GPRMC,111719.000,A,2838.0045,N,07713.3707,E,0.00,,120810,,,A*75,#01100101101010,6.5,0,0,12345.67,24.4,4.2,21,MCC,MNC,LAC,CellIDATL0×040×7A

# Protocol Description

## The protocol is having the following parts

- 0×20
- Starting character
- Signature of protocol
- IMEI number of device
- GPRMC strings
- Alerts
- Signature
- End character
- Checksum

- 0×20----Fixed(Space)
- 0×01/0×03 ----Starting Character of protocol

0×01—for live data and 0×03---for Memory data

- ATL ---Signature of the Protocol
- 356895037533745 - IMEI -15 CHAR

## THIS SECTION RELATES TO GPS DATA. $GPRMC STRING CONTAINS 13 FIELDS, WHICH ARE AS FOLLOWS:

- $GPRMC - GPS DATA -66 CHAR

$GPRMC,220516,A,5133.82,N,00042.24,W,173.8,231.8,130694,004.2,W,A*70

| Info | Description |
|---|---|
| $GPRMC | Fixed |
| 220516 | TIME STAMP (GMT) |
| A | VALIDITY A- VALID, V - INVALID |
| 5133.82 | CURRENT LATITUDE |
| N | NORTH/SOUTH(S) |
| 00042.24 | CURRENT LOGITUDE |
| W | WEST /EAST(E) |
| 173.8 | SPEED IN KNOTS |
| 231.8 | TRUE COURSE |
| 130694 | DATE STAMP (DDMMYY) |
| 004.2 | MAGNETIC VARIATION DEGREE |
| W | WEST/ EAST(E) of magnetic variation |
| A | Mode indicator, (A=Autonomous, D=Differential, E=Estimated, N=Data not valid) |
| *70 | CHECKSUM |

## • ALERTS

THIS SECTION RELATES TO VARIOUS ALERTS RAISED ON THE BASIS OF SYSTEM STATUS.

,#0110010011101,6.5,2345,0,12345.67 24.4,4.2,21,MCC,MNC,LAC,CellID

1 2 3 4 5 6 7 8 9 10 11 12 13

| Sl.No. | Value | Description |
|---|---|---|
| 1 | # | FIXED |
| 2 | 11011001001010 | STATUS OF VARIOUS I/O FROM I1 to I14 |
| 3 | 0 | RESERVED |
| 4 | 0 | RESERVED |
| 5 | 0 | RESERVED |
| 6 | 12345.90 | ODOMETER READING |
| 7 | 24.4 | ON-Board temp./Device Version |
| 8 | 4.2 | Device internal battery voltage |
| 9 | 21 | GSM signal Strength |
| 10 | MCC | Mobile country code[3 digits] |
| 11 | MNC | Mobile Network Code[3 digits] |
| 12 | LAC | Location area code(Mobile)[6 digits] |
| 13 | CellId | GSM cell ID[5 digits] |

Note:- In field number 7 there may be two different values in protocol.

1. Temperature of module OR
2. Firmware version _mobile no. ( if mobile number is not available, the IMSI number would come

- ATL -----------Signature
- 0×02/0×04 -------------end charactor

0×02 -----------for live data and 0×04 -----for memory data

- 0×7A----------check sum

Note:-

- The start character and end character are ASCII in protocol need to convert to hex.
- The checksum is ASCII in protocol need to convert to hex.
- If checksum is FF read it as 1B, due to some data sending limitation we cannot send 1B. So adjust the same at server end.

# Checksum Calculating Method :-

```
voidchksum_gprs(char * ptr)
{
unsignedint i = 0;
charchksum=0;
for(i=0;i<1000 && *ptr != 0x00;i++)
{
chksum ^= *ptr++;
}
return(chksum);
}
```

NOTE: FOR ALL DATA PACKETS, AT LAST YOU WOULD GET

CHECKSUM OF ALL BYTES IN PACKET BETWEEN STARTING CHARACTER AND END CHARACTER (INCLUDING BOTH STARTING CHARACTER AND END CHARACTER).

# Input Details

| INPUTS | Use | Description |
| --- | --- | --- |
| I1 | High Sense | Ignition, 0=OFF&1=ON |
| I2 | RESERVED | RESERVED |
| I3 | RESERVED | RESERVED |
| I4 | RESERVED | RESERVED |
| I5 | RESERVED | RESERVED |
| I6 | RESERVED | RESERVED |
| I7 | RESERVED | RESERVED |
| I8 | High Sense | Main Power(Fixed), 0=OFF and 1=ON |
| I9 | RESERVED | RESERVED |
| I10 | RESERVED | RESERVED |
| I11 | ARM/DIASRM | 1=ARM & 0=DISARM |
| I12 | Sleep Status | 1=Not in Sleep & 0= in Sleep |
| I13 | RESERVED | RESERVED |
| I14 | Movement Status | 1=In motion& 0= Not in motion |

# GPRS Commanding

The format for GPRS command would be

$MSG,Command&

| Sl.No. | Content | Discription |
|---|---|---|
| 1 | $MSG | Fixed header of command |
| 2 | Command | Command to device |
| 3 | & | End field |

# Sample Command

# GPRS command to check Version is

$MSG,VERSION<6906>&

**Note:-**

1. The protocol structure is for standard hardware, some hardware may not have the values for all field.
2. The E101 hardware supports only one 1-wire port, so either temperature or i-button would work at a time.
3. Please refer user manual of product for commands